

O' LEVEL PHYSICS

Practical Electricity

4.4

STUDENT NAME



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PHYSICS WITH
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SYLLABUS OUTLINE:

4.4 Practical electricity**4.4.1 Uses of electricity**

- 1 State common uses of electricity, including heating, lighting, battery charging and powering motors and electronic systems
- 2 State the advantages of connecting lamps in parallel in a lighting circuit
- 3 Recall and use the equation
power = current $\times$ voltage
 $P = IV$
- 4 Recall and use the equation
energy = current $\times$ voltage $\times$ time
 $E = IVt$
- 5 Define the kilowatt-hour (kWh) and calculate the cost of using electrical appliances where the energy unit is the kWh

4.4.2 Electrical safety

- 1 State the hazards of:
 - (a) damaged insulation
 - (b) overheating cables
 - (c) damp conditions
 - (d) excess current from overloading of plugs, extension leads, single and multiple sockets when using a mains supply
- 2 Explain the use and operation of trip switches and fuses and choose appropriate fuse ratings and trip switch settings
- 3 Explain what happens when a live wire touches a metal case that is earthed
- 4 Explain why the outer casing of an electrical appliance must be either non-conducting (double-insulated) or earthed
- 5 Know that a mains circuit consists of a live wire (line wire), a neutral wire and an earth wire and explain why a switch must be connected into the live wire for the circuit to be switched off safely
- 6 Explain why fuses and circuit breakers are connected into the live wire

USES OF ELECTRICITY:

HEATING:

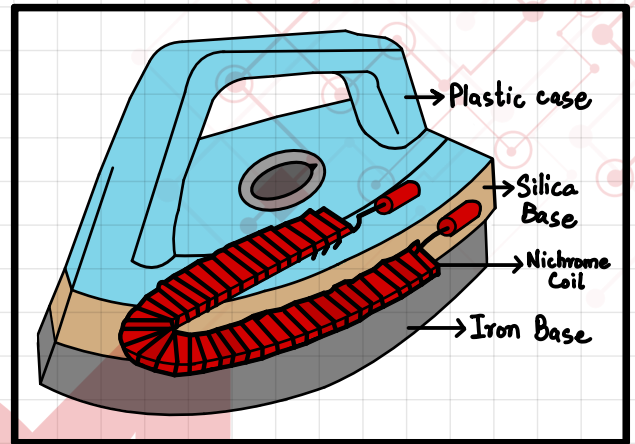
• ELECTRIC IRON:

We want maximum electrical energy to convert into thermal energy. For that:

- Nichrome wire is used (High resistance).
- Longer wire in the shape of coil ($\uparrow R \propto \uparrow L$).
- Thinner wire ($\uparrow R \propto \frac{1}{\downarrow A}$).

Coiled is placed over silica base which works as an electrical insulator as well as thermal conductor.

Coil heats up due to extreme resistance which in turn heats up silica base & silica base transfers heat to iron base.

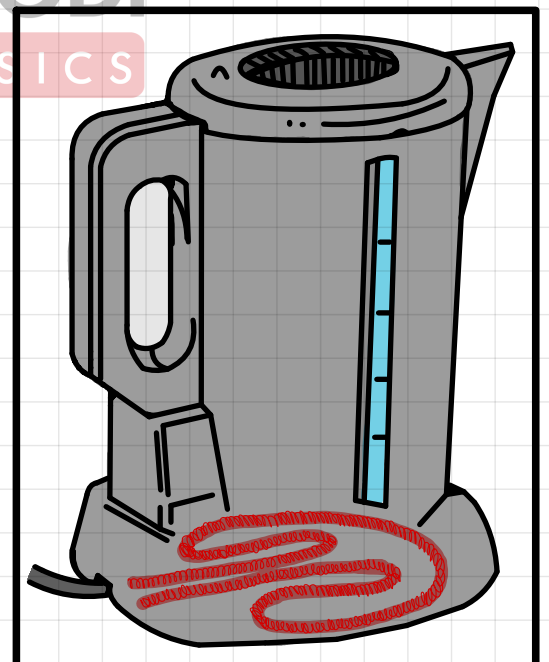


• ELECTRIC KETTLE:

- Nichrome wire is used (High resistance).
- Longer wire in the shape of coil ($\uparrow R \propto \uparrow L$).
- Thinner wire ($\uparrow R \propto \frac{1}{\downarrow A}$).

Coil heats up the tube due to radiation.

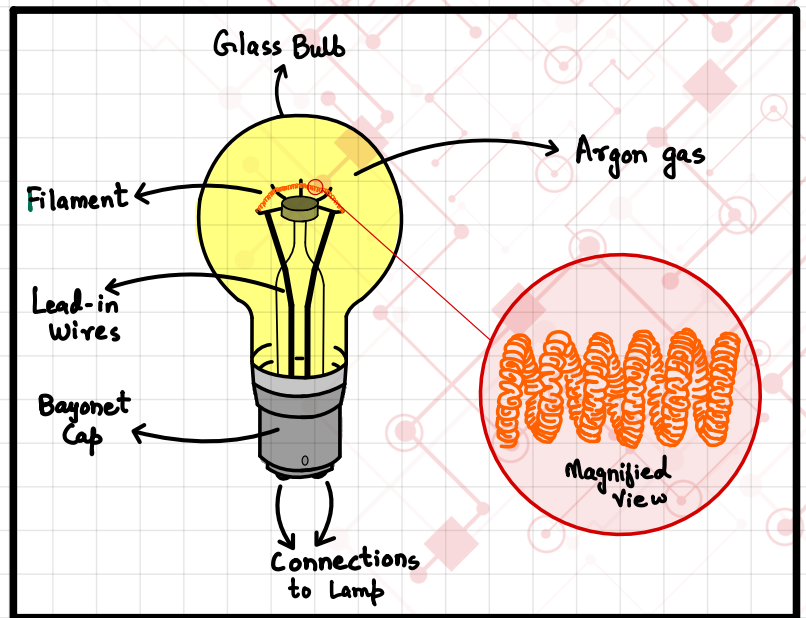
Water molecules make contact with the tube, becomes less dense, rises to the top. Cold water sinks to the bottom. This happens until thermal equilibrium is established.



LIGHTING:

FILAMENT LAMP:

- Tungsten wire is used (High resistance).
- Longer wire in the shape of coil ($\uparrow R \propto \uparrow L$).
- Thinner wire ($\uparrow R \propto \frac{1}{\downarrow A}$).

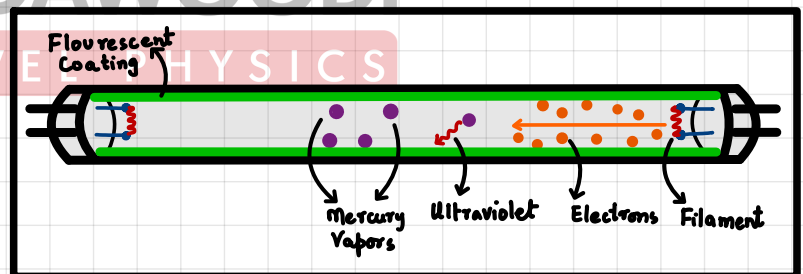


Tungsten converts electrical energy into heat energy due to high resistance & becomes red hot. While argon gas balances atmospheric pressure & doesn't let lamp's wall to shatter.

ADVANTAGES	DISADVANTAGES
• Low initial cost. • Cozy environment (low intensity lamps).	• 10% efficient. • Produces sharp shadows.

FLOURESCENT LAMP:

When electricity is provided to the electrodes, mercury vapours are energized & release ultra-violet light. Ultraviolet light strikes the flourescent powder & it glows.



ADVANTAGES	DISADVANTAGES
• Energy efficient (Low bill). • Soft shadows	• Expensive then filament lamp.

LED LIGHTS:

These are the most efficient lights which convert 40-50% of electrical energy into light energy.

They are made using semiconductors.

BATTERY CHARGER:

Battery charger transfer energy to a secondary rechargeable battery by forcing an electric current to flow in opposite direction to the normal current flow from the cell.

A battery charger should be switched off after battery gets fully charged so that battery donot get damaged due to overcharged.

POWERING MOTOR:

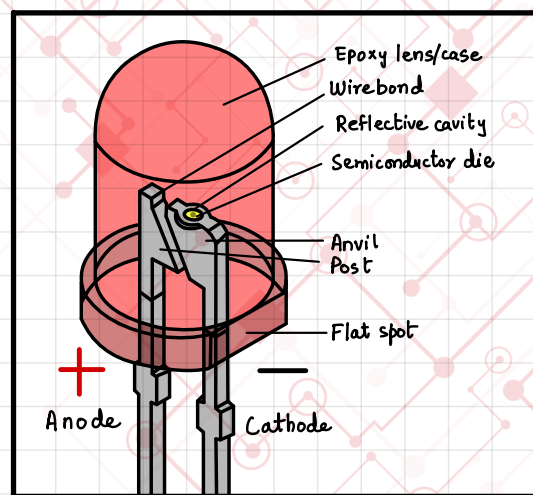
Motors are one of the most essential part of our daily life operations. It uses electrical energy & converts into kinetic energy.

CONNECTING LAMPS IN PARALLEL:

Every circuit is preferred to be connected in parallel with the mains supply i.e. across live & neutral wire & components receive full p.d. of mains supply (e.g: 230 V)

ADVANTAGES:

- P.d across each lamp is same so brightness of any lamp donot change.
- Each lamp can be switched ON/OFF independently.
- If one lamp fails to operate other will light up normally.



ELECTRICAL POWER:

"Product of current & voltage is called electrical power."

FORMULA:

Power = Current $\times$ Voltage

$$P = I V$$

Its S.I unit is watt (W). Commercial Unit kilo Watt (kW).

EXERCISE.

Q: An electric heater has a power rating of 2 kW . Calculate current if the supply voltage is 240 V .

$$\square \text{ kW} \rightarrow \square \times 1000 \text{ W}$$

$$P = I V$$

$$I = \frac{P}{V} = \frac{2000}{240}$$

$$I = 8.3 \text{ A}$$

ELECTRICAL ENERGY:

"Product of Power & time."

FORMULA:

Energy = Power $\times$ Time

$$E = P \times t$$

$$E = I V t$$

Its S.I unit is Joule (J). Commercial Unit kilo Watt hour (kWh).

EXERCISE.

Q: Electrical motor take 100 V voltage & draws a current of 3 A in 20 s time. Calculate the energy consumed by motor.

$$E = I V t = 3 \times 100 \times 20$$

$$E = 6000 \text{ J}$$

ELECTRICAL COST.

To calculate electrical cost, first convert power in kilowatt (kW) & time in hours (h). Then use formula of energy to calculate total no. of units (kWh). Finally multiply no. of unit with the unit cost.

EXERCISE:

$$\frac{100}{1000} = 0.1 \text{ kW}$$

$$\square \text{ s} \rightarrow \frac{\square}{3600} \text{ h}$$

Q: Lamp has a power rating of 100 W & is run for 2 hours.

Calculate the cost of running lamp if one unit cost is 20 Rs.

$$E = Pt = 0.1 \times 2$$

$$E = 0.2 \text{ kWh}$$

$$\text{Cost} = E \times 20 = 0.2 \times 20$$

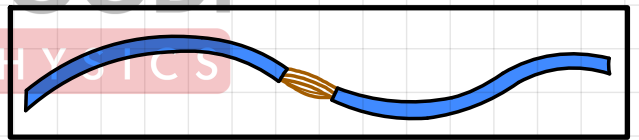
$$\text{Cost} = 4 \text{ Rs}$$

ELECTRICAL SAFETY:

HAZARDS:

• DAMAGED INSULATION:

Due to twisting, bending, curling the plastic insulation of cable damages which can cause electric shock.

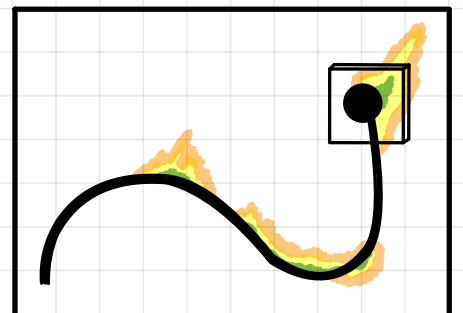


PRECAUTION:

Cover the damaged part using electrical tape or insulation sleeve.

• OVERHEATING CABLES:

Electric wiring becomes so hot due to excessive current which can even can fire. To reduce this, maximum current must be limited.



PRECAUTION:

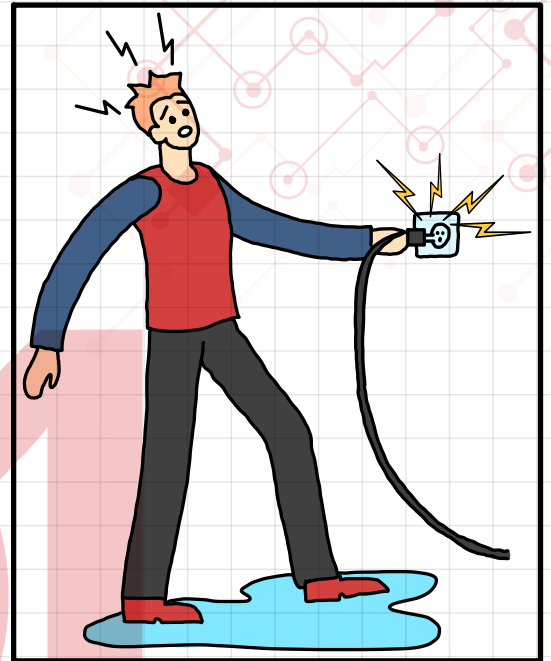
- Use fuse of correct ratings.
- Donot connect high current appliances like heater/air conditioner with thin wires. Use thicker wires.

DAMPED CONDITIONS:

Flow of current can be easy when skin is wet as wet skin has 10 times less resistance than dry skin. So chances of getting electric shock are high.

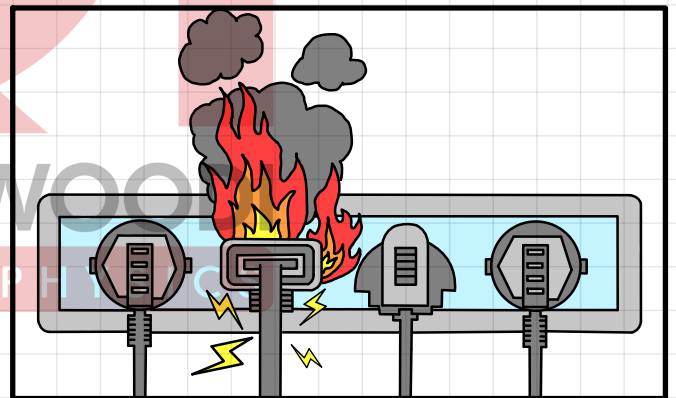
PRECAUTION:

- Wear insulating shoes with thick rubber.
- Approach electrical appliance when you are standing on dry floor.



OVERLOADED EXTENSION:

Overloaded extension can catch fire if it powers high current appliance as high current causes heating effect.



PRECAUTION:

Donot connect high current appliances with extension rather then use specially designed wiring for high currents.

Types Of Wires:

• LIVE WIRE

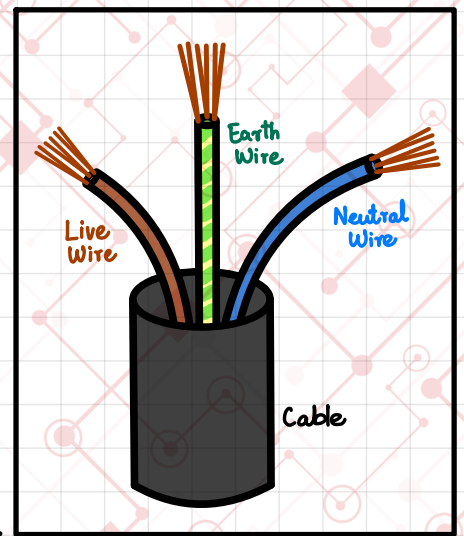
"A wire which is brown in colour & carries high voltage is called live wire."

• NEUTRAL WIRE:

"A wire which is blue in colour & has zero volts. It completes the circuit."

• EARTH WIRE:

"A wire which is yellow-green in colour & provides earth connection."

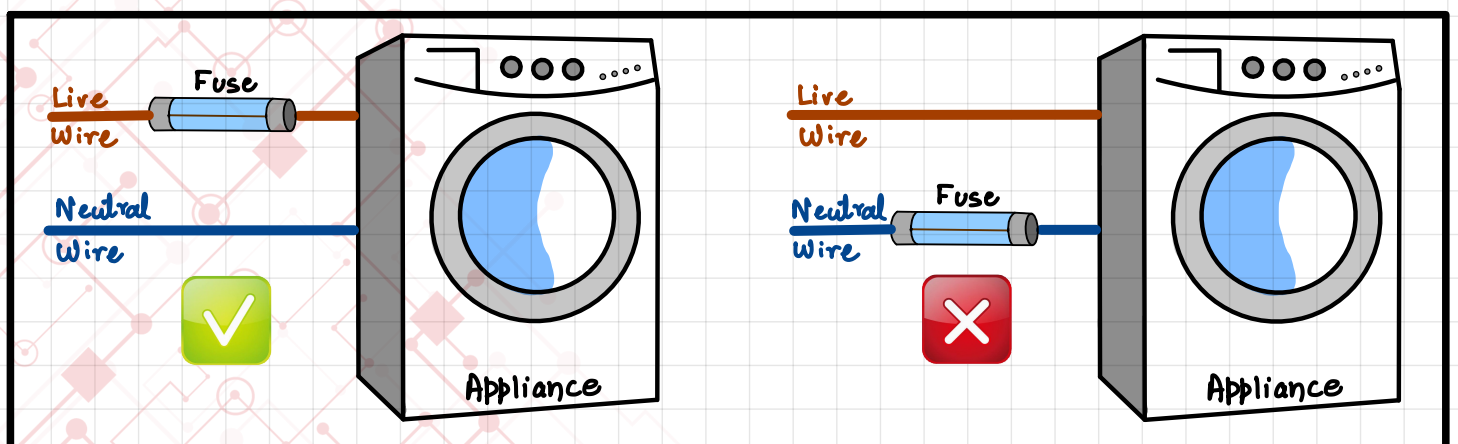
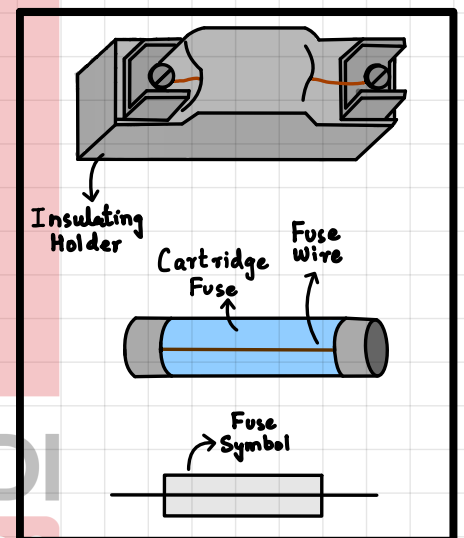


FUSE:

"Fuse is an electrical device which limits the amount of current in a circuit"

Fuse has a rating of how much it can allow current to pass through it. If current more than fuse rating flows through it, the wire inside it melts & disconnects the circuit from mains supply.

Fuse is always placed with live wire as when it blows, appliances can be completely isolated from high current.



Using high rating fuse more than the need of supply can cause large current flow which can cause fire.

FUSE RATING.

Q: Find out the fuse rating if appliance has a power rating of 1000W connected with mains 230 V supply.

• Find current:

$$P = IV$$

$$I = \frac{P}{V} = \frac{1000}{230}$$

$$I = 4.34 \text{ A}$$

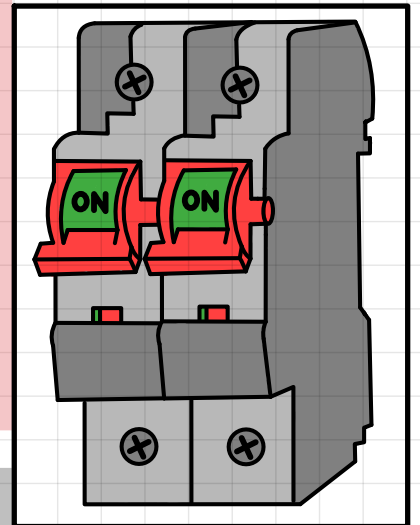
• Fuse rating should be slightly greater than current drawn.

0-5 A or 0-6 A.

TRIP SWITCH:

Trip switch also called circuit breaker are now used instead of fuse. They contain electromagnets inside them which break electrical contact when current becomes higher than rating. Circuit breaker operate much faster than fuse & can be used again & again by resetting the tripper.

It's rating is also chosen slightly higher than required current & is always placed with live wire.



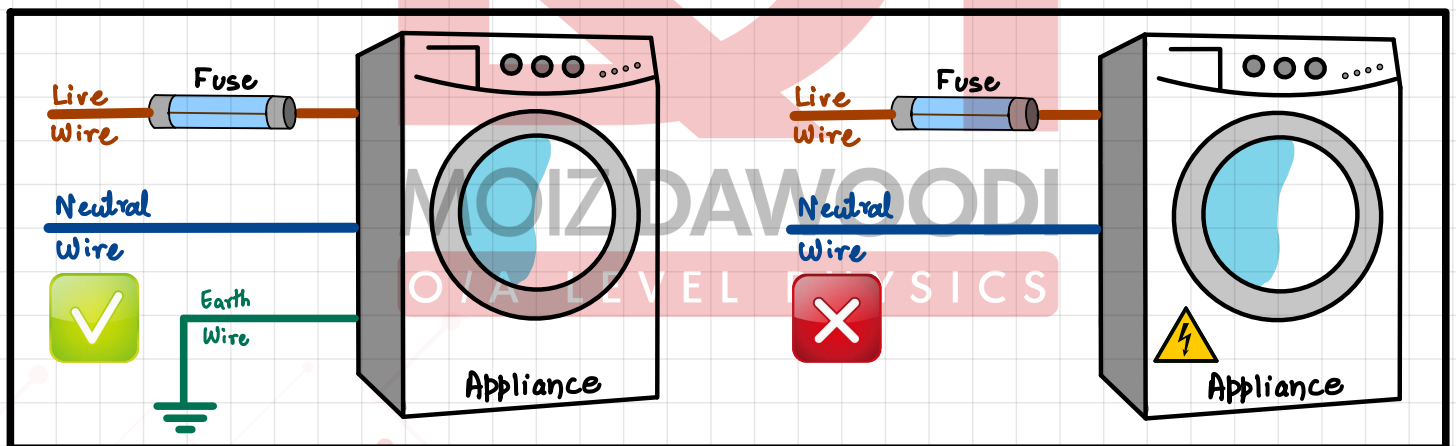
EARTHING: ($\perp$)

"Method of connecting charged metal with an earth connection in order to neutralise it."

PREVENTING ELECTRIC SHOCK USING EARTHING:

Due to damaged insulation when live wire touches metal casing, casing carries live current which can cause electric shock. To prevent shock, metal casing must be earthed so that all current goes to earth rather than through a person.

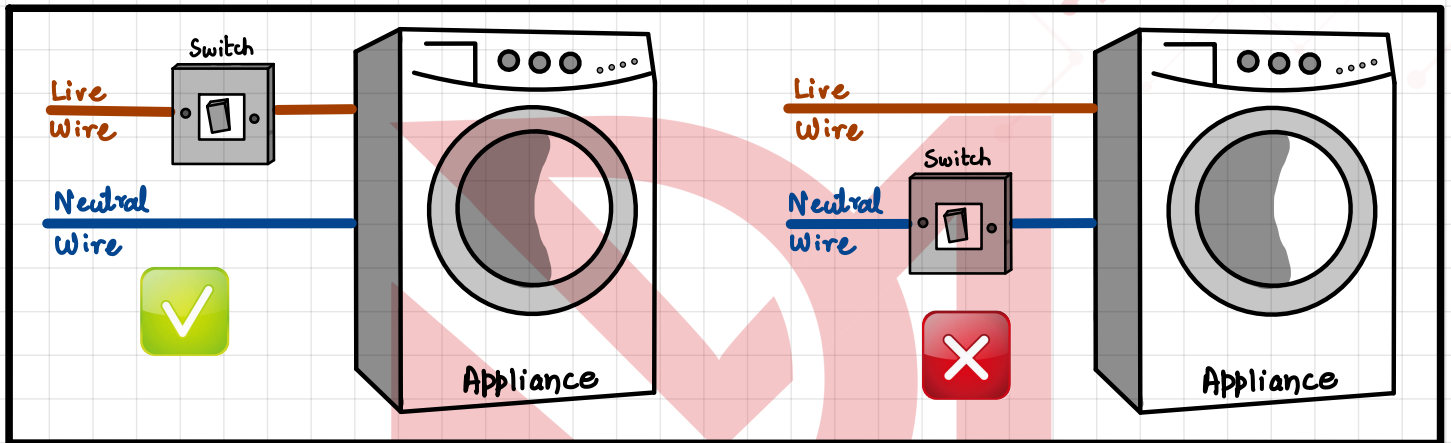
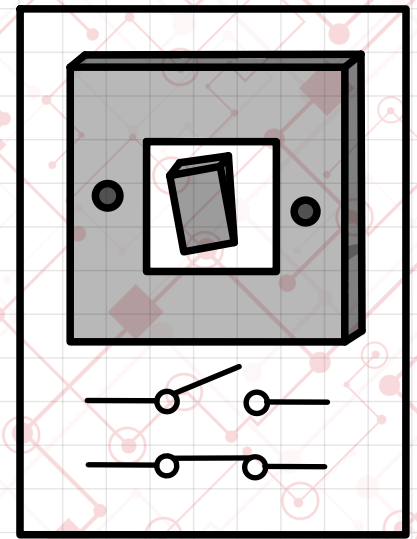
But this increases the amount of current flows from live to earth & due to heat wiring can catch fire. To prevent this fuse is also placed with live wire which melts when current exceeds the rating, so both appliance & person are safe.



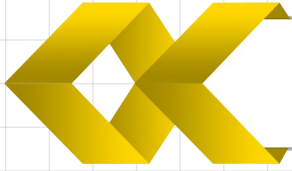
SWITCH:

"A component which makes the circuit ON/OFF is called switch."

It is always connected with the live wire as when it is OFF, appliance is completely isolated from current.



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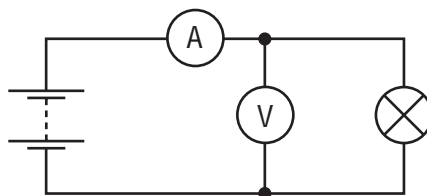


Match the following



High resistance wires	is always placed in live wire.
Electrical Power	is brown in colour & carries 230 V
Electrical Energy	are tungsten & nichrome wires.
Fuse	is product of current, voltage & time.
Earthing	doesn't get live as it is bad conductor.
Live wire	is product of current & voltage.
Rating of fuse	must be done with metal casing appliance.
Plastic	are used in home circuits.
Short circuit	should be slightly greater than current.
Parallel circuits	causes high current flows & causes fire.

01 The diagram shows a circuit.



The lamp is a **12W** lamp and is **working at normal brightness**.

$$P = IV$$

What **are the readings** on the meters?

	voltmeter reading / V ✓	ammeter reading / A ✓	
A	6	0.5	$= 3W \times$
B	12	0.5	$= 6 \times$
C	12	1	$= 12 \checkmark$
D	24	2	$= 48 \times$

[MJ2011/P11/Q31]

02 A lamp rated **6V, 2A** is switched on for **60s**.

$$E = IVt = 2 \times 6 \times 60$$

How much **energy** is used?

A 0.2 J

B 20 J

C 180 J

D 720 J

[MJ2011/P11/Q32]

03 The current in a filament lamp is **0.25 A** when working normally. The lamp is connected to a plug and the **mains a.c. supply**.

When the lamp is **switched on**, it **does not light**.

What is a **possible cause** for this?

A The earth wire in the plug is not connected. $\times$

B The fuse in the plug is 3A. $\times$

C The lamp only works on a d.c. power supply. $\times$

D The live wire in the plug is not connected.

[ON2011/P11/Q30]

04 Which appliance used on a 240 V supply is **most** likely to burn out its fuse?

	appliance	fuse
A	100 W lamp	1 A ✓
B	1 kW vacuum cleaner 1000 W	5 A ✓
C	2 kW fan heater 2000 W	3 A
D	3 kW electric fire 3000 W	13 A ✓

$$I = \frac{P}{V} \rightarrow \uparrow I \propto \uparrow P$$

$$\frac{100}{240} = 0.416 \text{ A}$$

$$\frac{1000}{240} = 4.16 \text{ A}$$

$$\frac{2000}{240} = 8.33 \text{ A}$$

$$\frac{3000}{240} = 12.5 \text{ A}$$

[ON2011/P11/Q31]

05 A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

$$E = Pt = 36 \times 10 = 360 \text{ J}$$

A 30 J

B 36 J

C 120 J

D 360 J

[MJ2012/P11/Q31]

06 What causes the fuse to blow in a mains electrical circuit?

A a person touches the live wire

B a person touches the neutral wire

C the live wire touches the earth wire

D the neutral wire touches the earth wire

[MJ2012/P11/Q32]

07 What is the current in the lamp?
A lamp is rated at 12 V, 600 mW.

$$I = \frac{P}{V} = \frac{600 \text{ m}}{12} = 50 \text{ mA}$$

A 20 mA

B 50 mA

C 2.0 A

D 5.0 A

[MJ2012/P12/Q31]

$$I = \frac{P}{V} = \frac{6000}{240} = 25 \text{ A}$$

08 A water heater uses 6 kW of electric power when connected to a 240 V circuit.
Which fuse is **most** suitable for use in this circuit?

A 5 A

B 13 A

C 30 A

D 50 A

[MJ2012/P12/Q32]

09 A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s? *Repeated*

A 30 J

B 36 J

C 120 J

D 360 J

[ON2012/P11/Q31]

10 An immersion heater is labelled 12 V, 60 W.

$$I = \frac{P}{V} = \frac{60}{12} = 5 \text{ A}$$

What is the current in the heater when connected to a 12 V supply?

A 0.20 A

B 5.0 A

C 12 A

D 60 A

[ON2012/P11/Q32]

$$I = \frac{P}{V} = \frac{60}{240} = 0.25 \text{ A}$$

- 11 A lamp is rated at 60 W on a 240 V supply.

What is the current in the lamp when used normally?

A 0.25 A

B 4.0 A

C 60 A

D 180 A

[ON2012/P12/Q31]

- 12 The cost of electricity is 16 c per kWh.

$$\text{Cost} = E \times 16 = Pt \times 16 = 3 \times 4 \times 16$$

What is the cost of using a 3 kW heater for 4 hours?

A $\frac{3 \times 4}{16}$ c

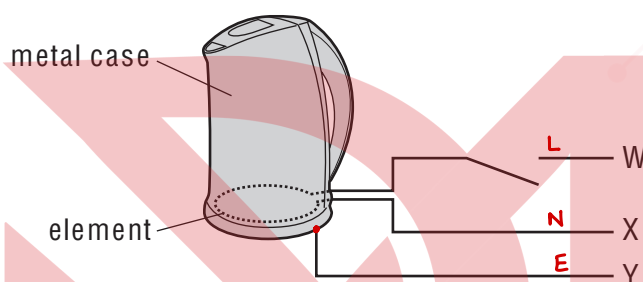
B $\frac{3 \times 16}{4}$ c

C $4 \times 3 \times 16$ c

D $4 \times 3 \times 60 \times 16$ c

[ON2012/P12/Q32]

- 13 The diagram represents part of a household circuit containing an electric kettle.



Which row correctly identifies the wires W, X and Y?

	W	X	Y
A	earth	live	neutral
B	live ✓	neutral ✓	earth ✓
C	live ✓	earth	neutral
D	neutral	live	earth

[MJ2013/P11/Q33]

- 14 Five electrical appliances are connected to the same socket and there is a very large current.

Why is this dangerous?

A The fuses blow in the appliances. ✗

B There is a greater risk of an electrical shock. ✗

C There is overheating in each appliance. ✗

D There is overheating in the socket.

[MJ2013/P12/Q36]

- 15 One kilowatt-hour of energy costs \$0.24.

How much does it cost to run a 2 kW heater for three hours?

A \$0.24

B \$0.48

C \$0.72

D \$1.44

[MJ2014/P11/Q32]

$$\text{Cost} = E \times 0.24 = Pt \times 0.24 = 2 \times 3 \times 0.24 =$$

- 16 An appliance uses a current of 3A.
Which row is correct for the fuse in this appliance?

	most suitable fuse rating / A	fuse connected in
A	5 ✓	earth wire
B	5 ✓	live wire ✓
C	13	earth wire
D	13	live wire ✓

[MJ2014/P12/Q33]

$$P = IV = 0.25 \times 240 = 60 \text{ W}$$

- 17 There is a current of 0.25A in a lamp connected to a 240V supply.
What is the input power to the lamp?

A 15 W B 60 W C 240 W D 960 W

[ON2014/P12/Q28]

- 18 Which unit measures the energy input to an electrical appliance?

A ampere ✗

B kilowatt-hour ✓ $E = Pt = \text{kWh}$

C volt ✗

D watt ✗

[ON2014/P12/Q29]

- 19 An electric lamp is marked 0.5A. It is connected to a socket marked '30A maximum'.
Which fuse is best to use in the lamp?

~~A~~ 0.5 A

B 3 A

~~C~~ 30 A

~~D~~ 40 A

[ON2014/P12/Q30]

- 20 When the flash on a camera is used, a charge of 1.5C flows through the flash-tube in 0.0030 s.
The average voltage across the flash-tube is 3600V.

What is the electrical energy supplied to the flash-tube and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400 ✓	1.8×10^6 ✓

$$P = IV = \frac{q}{t} \times V = \frac{1.5 \times 3600}{0.003} = 1.8 \times 10^6 \text{ W}$$

$$E = Pt = 1.8 \times 10^6 \times 0.003 = 5400 \text{ J}$$

[MJ2015/P11/Q31]

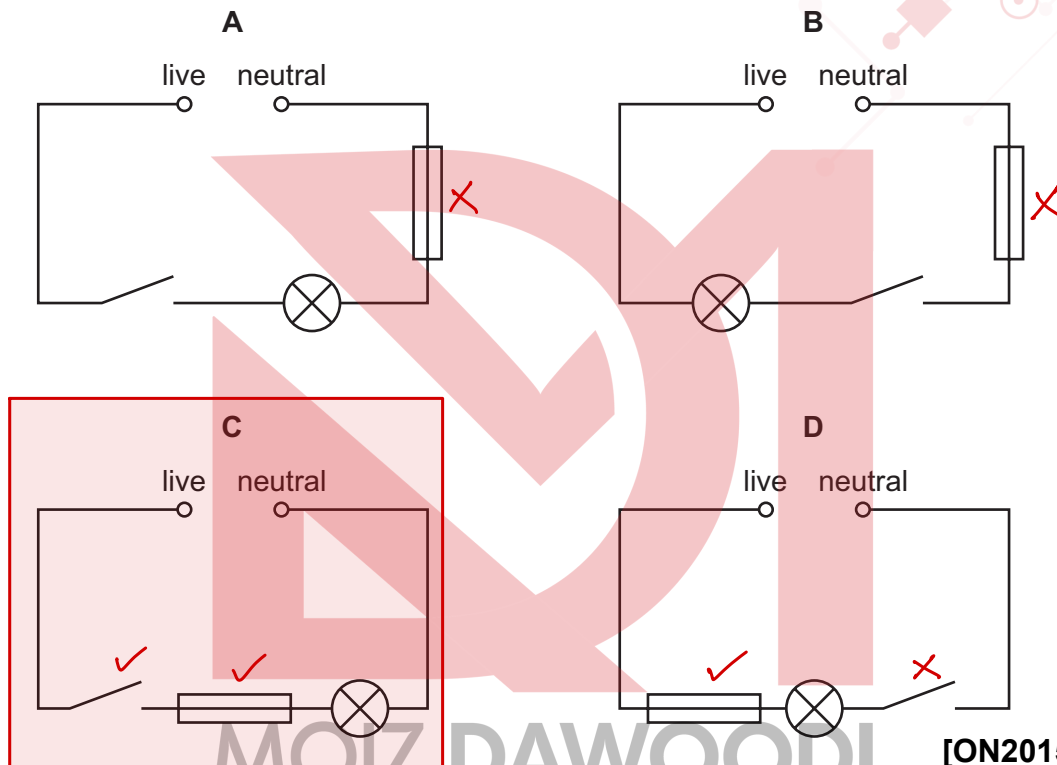
21 The diagram shows the information found on an electric kettle.

240 V	50 Hz
600 W	700 cm ³

What is the frequency of the electrical supply used to power the kettle?

- A** 50 Hz **B** 240 V **C** 600 W **D** 700 cm³ [MJ2015/P11/Q35]

22 Which diagram shows a lamp wired correctly to the mains supply in a house?



[ON2015/P11/Q31]

23 An immersion heater is labelled 24 V, 120 W.

$$I = \frac{P}{V} = \frac{120}{24} = 5 \text{ A}$$

What is the current in the heater when it is connected to a 24 V supply?

- A** 0.20 A **B** 5.0 A **C** 24 A **D** 120 A [ON2015/P12/Q34]

P 1

Mark Scheme

1	C
2	D
3	D
4	C
5	D
6	C
7	B
8	C
9	D
10	B

11	A
12	C
13	B
14	D
15	D
16	B
17	B
18	B
19	B
20	D

21	A
22	C
23	B

01 The cable from the mains plug to a washing machine contains a live wire, a neutral wire and an earth wire. The earth wire is connected to the metal case of the washing machine.

(a) Explain how connecting the earth wire to the metal case makes the washing machine safer.

Due to damaged insulation, metal case becomes live, all the current goes to the Earth leaving person touching it unharmed.

[2]

(b) When in use, the average input power to the washing machine is 500W.

$$\rightarrow \frac{500}{1000} = 0.5 \text{ kW}$$

$$\rightarrow \frac{45}{60} = 0.75 \text{ h}$$

Calculate the number of kWh of energy used by the washing machine in 45 minutes of use.

$$E = Pt = 0.5 \times 0.75$$

number of kWh = 0.375 kWh [2]

[MJ2011/P22/Q6]

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02 Fig. 2.1 is the circuit diagram of a fan oven that works from a 230 V a.c. mains supply.

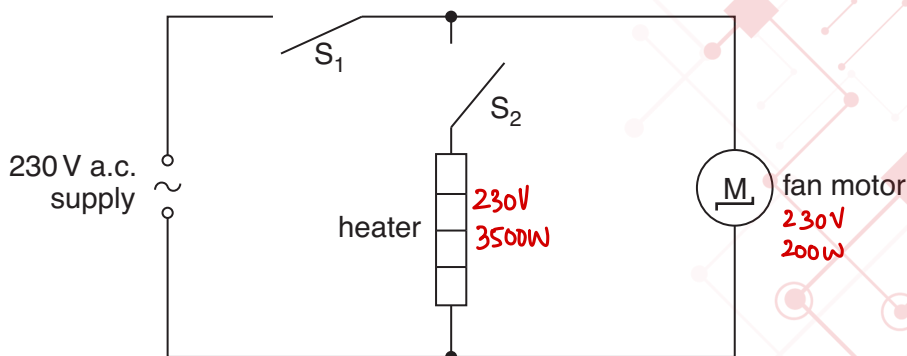


Fig. 2.1

The oven contains a 3500W electric heater, an electric fan operated by a 200W fan motor and two switches S_1 and S_2 .

- (a) (i) When switches S_1 and S_2 are open, both the heater and the fan motor are off, as shown in Fig. 10.2.

S_1	S_2	heater	fan motor
open	open	off	off
closed	open	off	on
closed	closed	on	on

Fig. 2.2

Complete Fig. 2.2 by writing “open” or “closed” in the boxes.

[2]

- (ii) Explain how the circuit represented by Fig. 2.1 ensures that the heater cannot be switched on when the fan motor is off.

For heater to be switched on S_1 must be closed which will eventually switch ON fan motor

[2]

- (iii) Suggest one reason why the circuit is designed so that the heater cannot be switched on when the fan motor is off.

So that oven may not be overheated.

[1]

- (b) (i) The heater and the fan motor are both switched on. Calculate the current supplied to the oven.

$$I = \frac{P}{V} = \frac{3500}{230} = 15.21 \text{ A}$$

$$15.21 + 0.869 = 16.079$$

$$I = \frac{P}{V} = \frac{200}{230} = 0.869 \text{ A}$$

$$\text{current} = 16.07 \text{ A} \dots\dots\dots [2]$$

- (ii) 1. Suggest a suitable rating for the fuse to be used with this oven.

$$0-18 \text{ A} \dots\dots\dots [1]$$

2. The fuse is placed in one of the three wires in the mains lead. State the wire in which it is placed.

$$\text{Live wire.} \dots\dots\dots [1]$$

- (iii) Explain how earthing the metal case of the oven makes it safer.

Due to damaged insulation, metal case becomes live, all the current goes to the Earth leaving person touching it unharmed.

.....
..... [3]

- (c) The cost of 1 kWh of electrical energy is 35 cents. Calculate the cost of leaving just the fan motor switched on for 12 hours.

$$E = Pt = \frac{200}{1000} \times 12 = 2.4 \text{ kWh}$$

$$\text{Cost} = E \times 35 \text{ c} = 2.4 \times 35 \text{ c}$$

$$\text{cost} = 84 \text{ c} \dots\dots\dots [3]$$

[ON2011/P21/Q10]

03 Fig. 3.1a shows a room heater. Fig. 3.1b is a diagram of the electric circuit of the heater.

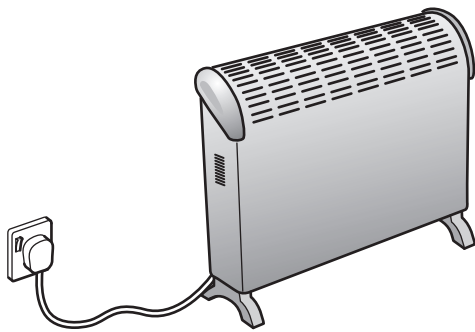


Fig. 3.1a

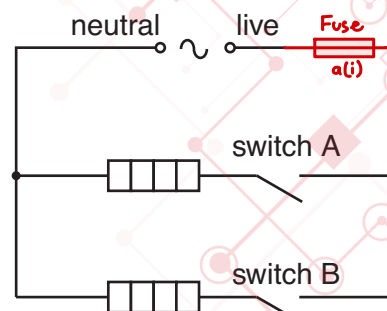


Fig. 3.1b

The fuse has not been drawn on the circuit diagram in Fig. 3.1b.

(a) (i) On Fig. 3.1b, draw the symbol for a fuse in the correct position. [2]

(ii) State the part of the room heater to which the earth wire is connected.

Metal case. [1]

(iii) The earth wire reduces the chance of an electric shock if a fault develops in the room heater.

1. State one fault that causes an electric shock when a person uses the room heater without an earth connection.

Due to damaged insulator, live wire makes contact with metal case making case live. [1]

2. Explain how using an earth connection prevents an electric shock.

Due to damaged insulation, current of live wire flows in the casing, if casing is Earthed, all current goes through earth wire leaving the person touching metal casing unharmed. [2]

- (b) (i) This type of room heater is **very efficient**. Explain **what this means**.

It converts maximum electrical energy to thermal energy.

.....[1]

- (ii) The room heater is a convector heater. Describe and explain how thermal energy (heat) passes around a room by convection.

Air near the heater becomes less dense & rises, while cold air sinks to the bottom. This continues until thermal equilibrium is established

.....

.....[3]

- (c) Fig. 3.2 shows the **power output** of the **room heater** when each **switch is closed**.

	power / W
switch A only closed	600
switch B only closed	
both switches closed	2100

Fig. 3.2

- (i) Determine the **power output** of the **room heater** when only **switch B is closed**.

$2100 - 600 = 1500$ power output = *1500 W* [1]

- (ii) The room heater is used with **both switches closed** for **2.5 hours**.

Calculate the **energy output** of the room heater

1. in **kilowatt-hours**,

$$E = Pt = \frac{2100}{1000} \times 2.5 =$$

energy = *5.25* kWh [2]

2. in **joules**.

$$E = Pt = 2100 \times 2.5 \times 60 \times 60$$

energy = *1.89×10^7* J [2]

[MJ2012/P21/Q9]

04 A 9.6 kW electric shower is powered by a 240 V mains supply.

- (a) (i) Calculate the electric current in the heating element of the shower.

$$I = \frac{P}{V} = \frac{9600}{240}$$

current = 40 A [3]

- (ii) Suggest an appropriate rating for the circuit breaker in the circuit for the shower.

0-45 A or 0-50 A [1]

- (b) A family uses the shower for a total time of 25 minutes every day. The cost of 1 kWh of electrical energy is 21 cents.

Calculate the daily cost of using the shower.

$$E = Pt = 9.6 \times \frac{25}{60} = 4 \text{ kWh}$$

$$\text{Cost} = E \times 21\text{c} = 4 \times 21$$

cost = 84 c [2]

[ON2013/P21/Q7]

05 An electric saw is rated at 1200 W. When working on a job that lasts for four days, a workman uses the saw, on average, for 75 minutes each day. The cost of 1 kWh of electrical energy is 21 cents.

- (a) Calculate the cost of using the saw for this job.

$$E = Pt = 1.2 \times \left(\frac{75 \times 4}{60} \right) = 6 \text{ kWh}$$

$$\text{Cost} = E \times 21\text{c} = 6 \times 21$$

cost = 126 c [3]

- (b) The metal case of the electric saw is earthed.

Explain how this protects the workman.

Due to damaged insulation, current of live wire flows in the casing, if casing is Earthed, all current goes through earth wire leaving the person touching metal casing unharmed. [2]

[ON2013/P22/Q7]

06 An electric hairdryer and an electric heater are connected to the mains supply, as shown in Fig. 6.1.

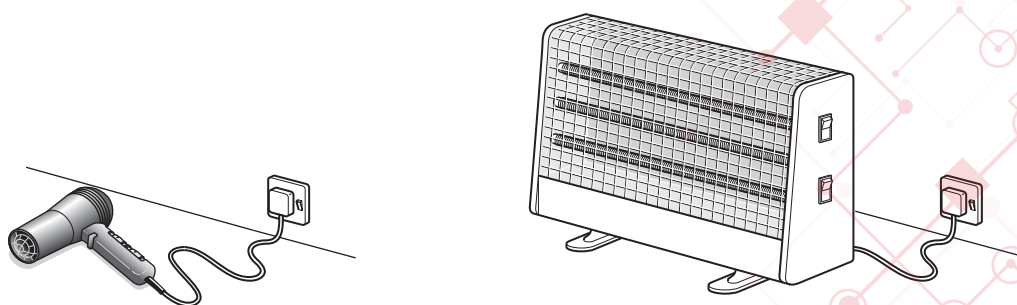


Fig. 6.1

The cable from the heater to the mains supply has a live, a neutral and an earth wire.

- (a) State the purpose of the neutral wire.

It completes the circuit.

[1]

- (b) The live wire in the electric heater touches the outer metal case.

Explain how the earth and the fuse together protect the user from electric shock.

Earthing provides alternative path to current in the metal casing to Earth leaving person unharmed. Since Earth has negligible resistance so current increases which surpasses the fuse rating & fuse blows, incompletes the circuit.

[2]

- (c) The hairdryer does not have an earth wire. Explain why this hairdryer is still safe to use.

It has plastic casing which does not get live.

[1]

- (d) In some modern homes, circuit breakers are used instead of fuses.

Suggest one advantage of using a circuit breaker rather than a fuse.

Can be used again & again.

[1]

[MJ2015/P22/Q7]

07 A spotlight in a television studio is at full brightness. The lamp inside the spotlight is powered by a 230 V supply and the current in the filament of the lamp is 27 A.

(a) Calculate the power of the lamp.

$$P = IV = 27 \times 230$$

power = 6210 W [2]

(b) The spotlight is kept switched on at full brightness.

(i) Calculate the energy transformed by the lamp in 30 minutes.

$$E = Pt = \frac{6210}{1000} \times \frac{30}{60} = 6.210 \times 0.5$$

energy = 3.105 kWh [1]

(ii) The cost of using one kilowatt-hour (kWh) of electricity is 23 cents.

Calculate the cost of using the spotlight for 30 minutes.

$$\text{Cost} = E \times 23 = 3.105 \times 23$$

cost = 71.41 c [2]

[ON2015/P22/Q8]

MOIZ DAWOODI
O/A LEVEL PHYSICS

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	22

- 6 (a) no shock // no electrocution B1
 (if) case becomes live // live touches case B1
 (b) correct conversion to kW, 0.5 seen // conversion to hours // $0.75 // \frac{45}{60} // (E =) P \times t$ C1
 0.375 // 0.38 // 0.37 (kW h) A1 [4]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – October/November 2011	5054	21

- 10 (a) (i)
- | | | | |
|--------|--------|--|--|
| | | | |
| | | | |
| closed | open | | |
| closed | closed | | |
- B1
B1
- (ii) S_1 closed → motor on B1
 S_1 open → heater off B1
- (iii) the heater would overheat/burn/melt or more efficient cooking/circulation described B1
- (b) (i) $(I =) P/V$ or 3700/230 or 3500/230 or 200/230 or 15(.217) C1
 16(.08695)A or 16.1 A A1
- (ii) integral value: 17 → 40 A or up to candidate's (b)(i) + 24 live B1
 B1
- (iii) if case becomes live or if live wire touches case B1
 fuse blows B1
 live/supply disconnected/case safe to touch/prevents shock/prevented electrocution B1
- (c) 0.20/3.5/3.7 (kW) or $200/3500/3700 \times 12 \times 35$ C1
 0.20/3.5/3.7 × 12 × 35 or 1470 c or 1554 c or 84000 c C1
- 84 c or \$0.84 (allow €/£/R etc.) A1 [15]



Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2012	5054	21

- 9 (a) (i) fuse symbol correct
in live wire before junction of two elements B1
B1
- (ii) the (metal) case/outside B1
- (iii) 1. live wire touches case; live touches person B1
2. current goes to earth; current does not go through the person
fuse blows B1
B1
- (b) (i) most of the energy output is useful/heat; little energy is wasted; B1
- (ii) hot air rises (**not** heat rises) B1
density of hot air is lower B1
convection current mentioned OR hot air rises and cold air falls B1
- (c) (i) 1500 W B1
- (ii) 1. conversion to kW seen on any power; 2.1 (kW) seen C1
5.25; 5.2; 5.3 (kWh) A1
2. $E = P \times t$ in any form, algebraic or using any power or time e.g. 600×2.5 ,
 600×150 C1
 1.89×10^7 (J) OR $3.6 \times 10^6 \times$ (c)(ii)1. A1 [15]

Page 3	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5054	21

- 7 (a) (i) $(I =)P/V$ or $9.6/240$ or 9600 C1
 $9600/240$ or 0.040 C1
40 A A1
- (ii) any whole number from 41 to 99 (incl.) with unit (A) B1
(e.c.f. from 0.040 A: 1,2,3 A)
- (b) $9.6 \times 25 \times 21$ or $9.6 \times 25/60$ or $9.6 \times 25/60 \times 21$ or 5040 c or \$50.40 etc. C1
84 c or \$0.84 or €0.84 or £0.84 or Rs0.84 etc. (85.7/86c from 0.42h) A1 [6]

O / A LEVEL PHYSICS

Page 4	Mark Scheme	Syllabus	Paper
	GCE O LEVEL – October/November 2013	5054	22

- 7 (a) a power $\times$ a time $\times$ the unit price
 (e.g. $1.2 \times 75/60 \times 4 \times 21$ or $1200 \times 75/60 \times 4 \times 21$ or $1.2 \times 75 \times 4 \times 21$
 or $1.2 \times 75/60 \times 21$ or 5 (hr) or 6 (kW h)) C1
 a power $\times$ a time $\times$ the unit price and with maximum of one physics
 error (i.e. use of 1200 or omits 60 or omits 4)
 (e.g. $1200 \times 75/60 \times 4 \times 21$ or $1.2 \times 75 \times 4 \times 21$ or $1.2 \times 75/60 \times 21$ or
 126 000 or 7560 or 31.5 (accept 0.21 for 21 and 75.60 and 0.315)
 (if this C mark is scored so is the previous one)
 126/130 c or \$1.26/1.30 or €/£/Rs 1.26/1.30 etc. A1
- (b) (if) case becomes live or live wire touches the case B1
 fuse blows or (large) current to earth or no current in workman
 (ignore excess; not “some current”) B1 [5]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	22

- 7 (a) to provide a complete circuit (with live) B1
 or to pass current back to mains
 or provide a return path for the current
- (b) current/charge/electrons flow to earth/earth wire/ground (when live touches
 case) B1
 fuse melts/blows and disconnects circuit/cuts live/stops current B1
- (c) doubly insulated B1
 or case/body made of plastic/insulator/not made of metal
 or user cannot touch metal
- (d) (circuit breaker) B1
 • turns off/acts fast(er)
 • can be reset
 • easy to see it has tripped/switched
 • can detect small difference between live and neutral currents / small
 (leakage) current to earth

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2015	5054	22

- 8 (a) ($P =$)VI or 230×27 C1
 6200/6210W or 6.21/6.2kW A1
- (b) (i) $1.1/1.12/1.1178 \times 10^7$ J B1
- (ii) $6.21 \times 0.5 \times 23$ or $6.21 \times 30 \times 23$ or $3.1/3.105 \times 23$ C1
 71/71.3/71.4/71.415 c or \$ 0.71/0.714/0.71415 A1 [5]